

FROM SUPERVISED TO UNSUPERVISED LEARNING FOR LAND COVER ANALYSIS OF SENTINEL-2 MULTISPECTRAL IMAGES.

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ABSTRACT

Sentinel-2 provides a large volume of the multi-spectral multi-resolution dataset. Training deep convolutional architecture with such a dataset is still a challenging task in land cover classification due to the absence of ground truth. Also, the selection of appropriate deep architecture for handling the Sentinel-2 dataset is another challenging task. In this paper, we propose a convolutional neural network (CNN) architecture to extract the information from various combinations of bands in Sentinel-2 imagery. We use a loss function, proposed in an earlier work, to train our model in an unsupervised manner. Recent advances in deep learning allow for transferring knowledge from one data set to another. Thus, in our study, we aim at analyzing the “transfer learning” capabilities of our proposed network to land cover classification in Sentinel-2 images. Pre-trained weights of a deep neural architecture, which is trained with very high-resolution optical satellite imagery from *Aviris* is transferred to a network of almost similar architecture for processing Sentinel-2 data. We used *Salinas*, a publicly available hyper-spectral dataset, to train this architecture in a supervised fashion and, finally, use transfer learning and fine-tuning on the architecture handling Sentinel-2 data for clustering. Experiments show that bands of 60m resolution have a positive combining effect with bands of 20m resolution for segregating waterbody, urban settlement, and tree canopies distinctly.

Index Terms— Convolutional neural network, Land-cover classification, Clustering, *Aviris* Hyperspectral images, Sentinel-2 Multispectral Images.

1. INTRODUCTION

Land-cover classification plays a vital role in many applications such as land use monitoring, urban ecosystem service mapping, resource assessment, and environmental protection [1, 2]. However, manual study is not feasible in many situations. Therefore, it becomes of paramount importance to develop automated land cover classification algorithms using remote sensing systems. With the launch of the Sentinel-2 program, we have a substantial resource of data motivating us to explore deep learning for their application. Currently, deep neural networks are found to be highly effective in image

classification due to its strong capability to depict high-level and semantic aspects of images. Still, their utility in remote sensing domains is less visible. One of the challenges in using these data for a task like supervised classification is to obtain ground truths. Thus, unsupervised learning becomes an attractive choice in analyzing such data. It is challenging to train a deep convolutional neural network (CNN) in an unsupervised manner. In this paper, we propose a new CNN architecture for landcover analysis with Sentinel-2 images, in the absence of any labeling information. We adopt transfer-learning [3] from a neural architecture trained with supervised labels of a publicly available *Salinas* (hyperspectral image) dataset. A patch-wise clustering is initially conducted on Sentinel-2 image by exploring multi-scale contextual information extracted by our network. We investigate the impact of the combination of bands that belong to different resolutions in the classification process. In summary, the contributions of this paper are as follows:

1. We propose a new layer in deep convolutional architecture, which enables us to achieve land-cover classification by using various combinations of sentinel-2 bands of different resolution.
2. We present a scheme to use “multi-scale filter bank” in the feature extraction layer to extract appropriate features from patches.
3. We adopt a novel strategy of applying how transfer learning in the current context. We have shown weights learned by the hyperspectral image in the proposed network become useful to train the system in an unsupervised manner for Sentinel-2 multispectral images.

The remainder of the paper is organized as follows: In Section 2, we describe datasets used in our study. In Section 3, the contribution of this paper is presented. In Section 4, methodologies adopted for implementation are discussed. We present the results of experiments in Section 5. Finally, we conclude our work in Section 6.

2. DATASET

The Sentinel 2 Level-1C product provides per-pixel radiometric measurements in Top Of Atmosphere (TOA) reflectances with all parameters to transform them into radiances. Level-1C products are resampled with a constant Ground Sampling

Distance (GSD) of 10m, 20m, and 60m depending on the native resolutions of different spectral bands. We have removed the *cirrus* band from our study. Table 1 depicts briefly information about them. The area under study along with their geographical centres [latitude, longitude], includes Medinipur [22°25'21.82" N, 87°18'28.29" E] district of West Bengal, India. They cover water bodies, forests, agriculture regions, tree canopies, and urban settlements. To use transfer

Table 1: Specifications of the Multi Spectral Instrument (MSI) on the Sentinel-2 satellite system.

Spectral Band	center wavelength	spatial resolution	Spectral Band	center wavelength	spatial resolution
B1	443	60	B7	783	20
B2	490	10	B8	842	10
B3	560	10	B8A	865	20
B4	665	10	B9	945	60
B5	705	20	B11	1610	20
B6	740	20	B12	2190	20

learning, we have selected the hyperspectral *Salinas* dataset, widely used by researchers. Table 2 describes the type of classes present in the *Salinas* dataset.

Table 2: Ground-truth classes present in the *Salinas* images

#	Class	#	Class
1	Brocoli_green_weeds_1	9	Soil_vinyard_develop
2	Brocoli_green_weeds_2	10	Corn_senesced_green_weeds
3	Fallow	11	Lettuce_romaine_4wk
4	Fallow_rough_plow	12	Lettuce_romaine_5wk
5	Fallow_smooth	13	Lettuce_romaine_6wk
6	Stubble	14	Lettuce_romaine_7wk
7	Celery	15	Vinyard_untrained
8	Grapes_untrained	16	Vinyard_vertical_trellis

3. OUR CONTRIBUTION

3.1. Network architecture

The proposed network architecture consists of mainly three blocks, namely, band-combination, feature extraction, and classification.

3.1.1. Band-combination

We assume bands with different resolutions have a disparate impact on land cover classification. If we extract features individually from bands of the same resolution, it will increase the number of network parameters to a great extent. To contain this, we enforce band combination to make the number of parameters in the deep network small. Initially, we re-sample bands of 20m and 60m resolutions to 10m resolution using the nearest neighbor up-sampling technique in the SNAP toolbox. We stack bands that come from the same resolution initially. We propose a linear layer, namely, a weighted sum of

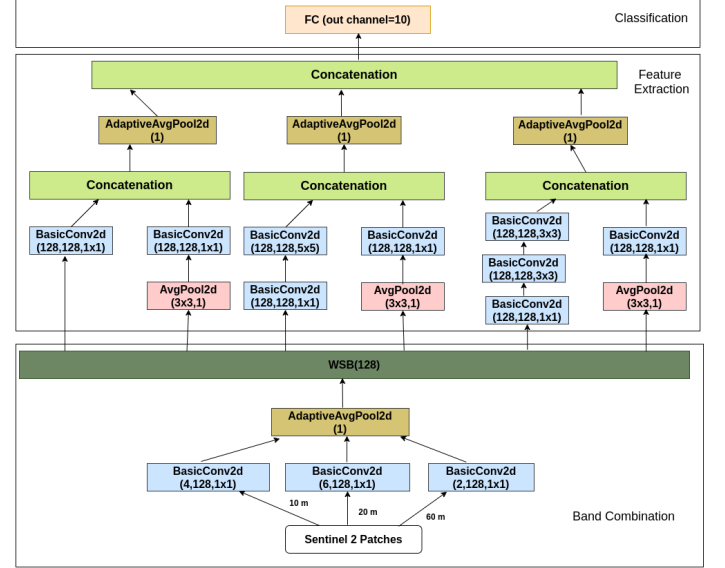


Fig. 1: Diagram of proposed network

bands (WSB) to serve the purpose of band-combination. This layer has three learnable parameters a , b , and c , and it follows Equation (1). We use the notation X_{10} , X_{20} , and X_{60} to refer stacked bands which have resolution 10m, 20m, and 60m before re-sampling.

$$Y = aX_{10} + bX_{20} + cX_{60} \quad (1)$$

We first pass X_{10} , X_{20} , and X_{60} separately to “BasicConv2d” layer followed by adaptive average pooling (AdaptiveAvgPool2d) layer. Each “BasicConv2d” layer in Figure 1 contains a 2-D batch-normalization layer followed by the rectified linear units (ReLU) after 2-D convolution layer. The parameter of the 2-D convolution layer is the input channel, output channel, and kernel size. Each rectangular block of “BasicConv2d” in the diagram contains parameters of the 2-D convolution layer. We denote this by $\mathbb{C}_{k_1 \times k_1 \times B}$, where k_1 refers to the kernel size of the convolution layer and B is the number of the input channel.

3.1.2. Feature-Extraction

Intuitively a “multi-scale filter bank” is capable of extracting more appropriate features from small patches. Hence, we adopt inception [4] like module for feature extraction. There are three inception architectures in Figure 1. Each of them has two parallel flows of processing. In one of them, it contains a series of “BasicConv2d” layers with varying sizes of the kernel, and the other one has “AvgPool2d” followed by a “BasicConv2d”. Each block of “AvgPool2d” depicts the kernel size and the stride value for the average pooling layer in the diagram. We denote this by $\mathbb{P}_{k_2 \times k_2}$, where k_2 refers to the kernel size of the pooling layer.

3.1.3. Clustering

Outputs of the feature extraction block are flattened and fed to the fully connected layers whose output channel is the number of classes. The fully connected layer is followed by a 1-D Batch-Normalization (BN) layer and a softmax activation function. In general, the classification layer can be defined as

$$p = \text{softmax}(\text{BN}(Wx + b)) \quad (2)$$

Where, W is the weight matrix of fully connected layer, x is the input to this layer and b is the bias. In Figure 1, FC represents fully connected layers followed by a 1-D Batch-Normalization layer. The *out channel* represents the number of output nodes in a fully connected layer. We set the output channel to the desired number of classes. In our experiment, we set it to 10. For unsupervised learning output nodes of classes to be considered as nodes of clusters.

3.2. Transfer learning

We find that the loss function used in [5] is not efficient to train the model from scratch for land cover clustering with Sentinel-2 datasets. Hence, we train our model with publicly available *Salinas* hyperspectral dataset with their labeled classes. In this case, the number of clusters in unsupervised learning is kept the same as the number of classes. The loss function used for supervised training is the cross-entropy function, whereas for unsupervised learning, it is in the form of a function as reported in [5]. To train with hyperspectral dataset, we replace the band combination block with one “BasicConv2d” layer for this purpose. All bands in the hyperspectral image have same spatial resolution. Therefore, we have modified the architecture. During training, we copy the weights of this “BasicConv2d” layer into each of three “BasicConv2d” layers in Figure 1 in the band combination layer.

4. METHODOLOGIES

4.1. Data normalization and patch extraction

We normalize each of the twelve bands using Equation (3) after re-sampling all the bands into 10m resolution.

$$\frac{X - \min(X)}{\max(X) - \min(X)} \quad (3)$$

Where X represents one of the twelve bands and $\min(X)$ and $\max(X)$ represents minimum and maximum of all the pixel intensities of that band X , respectively. We form a set of triplets of patches (10m, 20m, and 60m resolution) with no overlap (stride 8). The dimension of the triplet is $[4 \times 8 \times 8, 6 \times 8 \times 8, 2 \times 8 \times 8]$.

4.2. Model training

The successful employment of the loss function [5] requires the size of mini-batch is as large as possible. The task of the loss function is to group patches into K bins where K is the number of classes. For a good clustering, each cluster needs to be homogeneous, and the distance between any two clusters must be high. Therefore, it is necessary to have at least a few representatives of each class in the mini-batch. To ensure the above fact, we kept the size of the mini-batch is as large as possible. For our experiment, we consider the size of the mini-batch is 2000. We have used Adam optimizer with a learning rate of 0.001 and a weight decay of 0.001 to train the proposed model. We train our proposed model for 50 epochs.

5. RESULTS AND DISCUSSIONS

Table 3: Impact of different combinations of bands for land cover classification

Model	Initial [a,b,c]	Final [a,b,c]
M1	[0.7, 0.2, 0.1]	[0.52, 1.73e-09, -3.38e-11]
M2	[0.3, 0.3, 0.3]	[-5.47e-11, 0.54, -2.24e-09]
M3	[1.0, 1.0, 1.0]	[0.44, -4.19e-08, 0.50]
M4	[0.1, 0.3, 0.6]	[-2.33e-06, 0.30, 0.55]

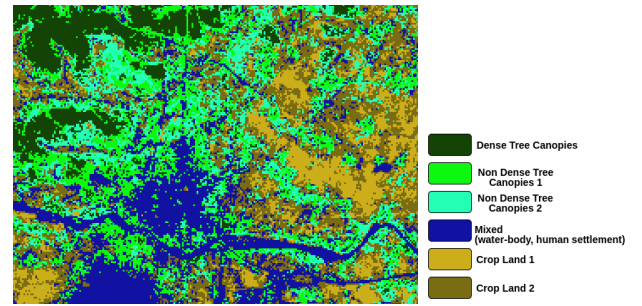


Fig. 2: Land cover Segmentation by model M1.

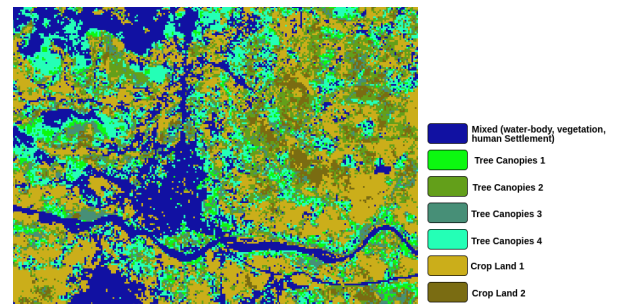


Fig. 3: Land cover Segmentation by model M2.

Figure 2 and Figure 3 depict the land cover classification map, which has the dominant effect of either band of 10m or 20m. We have verified our result consulting the toposheet (N0. F45J7)¹. We have converted the toposheet into a Geo-TIFF image. We use *QGIS* to project our land cover map on the toposheet to validate our work qualitatively. We find that the model M1 categorizes tree canopies broadly into two categories: dense and non-dense. Also, it categorizes croplands into two groups. It has identified waterbody and human settlement into the same group. On the other hand, model M2 groups waterbody, dense tree canopies, and built-up into one group. Also, M2 categorizes tree canopies into four groups and croplands into two.

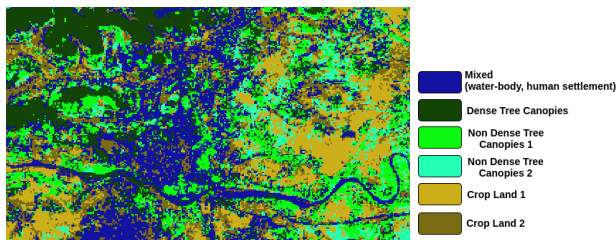


Fig. 4: Land cover Segmentation by model M3.

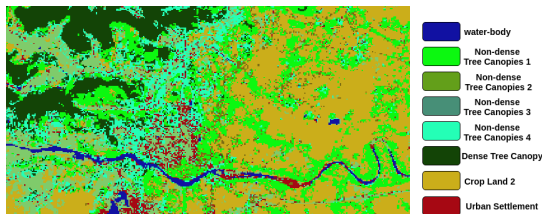


Fig. 5: Land cover Segmentation by model M4.

Our experiments reveal that combining bands of 60m resolution with either 10m or 20m resolutions shows improvement in classification. But they can not alone give any proper land cover segmentation. In fact, Figure 5 shows that M4 segregates waterbody and human settlement distinctly. But, river sand bed gets misclassified into human settlement. M3 and M4 classify tree canopies into two categories: dense and non-dense. However, non-dense canopies are clustered into four sub-clusters by M4 and two sub-clusters by M3. M4 identifies cropland into a single cluster, whereas M3 finds two subclusters in it.

6. CONCLUSION

In this paper, we propose a deep convolutional neural network which investigates the behavior of different combination of various bands in Sentinel-2 for land cover classification. Besides, we explore the capabilities of the process of “transfer

learning” to adopt a pre-trained (using supervised learning) network from high-resolution optical *Aviris* imagery, to be applied to satellites with lower geometric resolution such as Sentinel-2 in an unsupervised manner. Our results show that our model pre-trained with hyperspectral images can produce meaningful land cover clusters in Sentinel-2. Finally, the parameters of the proposed linear layer (*WSB*) reveals that combining the effect of bands of 60m and 20m resolution is effective. Waterbody and human settlement are identified as two different classes in such a combination. These two classes have a tendency to appear as a single class in multispectral images [6]. However, the quality of land cover clustering depends upon the initial parameters of the *WSB* layer of the network. The automation of getting an appropriate combination of various bands may be considered in a future study.

7. REFERENCES

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¹collected from ‘Geological Survey of India, 27, Jawaharlal Nehru Rd, New Market Area, Kolkata, West Bengal 700016’